

The partially bicyclic semigroup S_2 has (F_{*s})

Literature-implied answer for arXiv:1207.4531

11 August 2026

Status. Literature-implied affirmative answer (full answer to Question 3). The implication below combines the presentation of S_2 with a later nonlinear Ryll–Nardzewski theorem. It is not presented as a new theorem of the run.

Source question. Lau and Zhang, *Fixed point properties for semigroups of nonlinear mappings and amenability*, arXiv:1207.4531, ask in Question 3 on printed page 34:

Does the partially bicyclic semigroup

$$S_2 = \langle e, a, b, c : ab = ac = e \rangle$$

have the fixed-point property (F_{*s}) ?

Here (F_{*s}) says that every norm-nonexpansive representation of the discrete semigroup on a nonempty norm-separable weak-star compact convex subset Q of a dual Banach space, with weak-star continuous action, has a common fixed point. Their convention is $T_{st} = T_s \circ T_t$.

Decisive supporting theorem. Wiśnicki, *Around the nonlinear Ryll–Nardzewski theorem*, arXiv:1903.12123v2, proves in Corollary 3.3 on printed page 9 that if Q is a norm-separable weak-star compact convex subset of a dual Banach space, then every weak-star continuous, norm-nonexpansive, norm-distal semigroup action on Q has a common fixed point [2]. (The supporting paper writes “separable weak-star compact” after explicitly identifying the relevant hypothesis as norm separability.)

Theorem. *The partially bicyclic semigroup $S_2 = \langle e, a, b, c : ab = ac = e \rangle$ has property (F_{*s}) .*

Proof. Consider an arbitrary representation covered by (F_{*s}) , acting on Q , and put

$$A = T_a, \quad B = T_b, \quad C = T_c.$$

The defining relations and the representation convention give

$$AB = I_Q = AC.$$

For all $x, y \in Q$, nonexpansiveness gives

$$\|x - y\| = \|ABx - AB y\| \leq \|Bx - B y\| \leq \|x - y\|.$$

Thus B is an isometric embedding of Q into itself. The identical argument with C shows that C is an isometric embedding.

Let R be the semigroup of self-maps of Q generated by B and C . Every member of R is a composition of isometric embeddings and is therefore an isometric embedding. Consequently the R -action is norm-nonexpansive and norm-distal: for $x \neq y$,

$$\|Tx - Ty\| = \|x - y\| > 0 \quad (T \in R).$$

Every member of R is weak-star continuous, because B and C are. Wiśnicki's Corollary 3.3 therefore supplies $q \in Q$ fixed by all of R , in particular

$$Bq = q = Cq.$$

The relation $AB = I_Q$ now yields

$$Aq = ABq = q.$$

Thus q is fixed by the three generators a, b, c , and hence by every element of S_2 . Since the original representation was arbitrary, S_2 has (F_{*s}) . $\square$

Provenance and scope. The supporting paper cites Lau–Zhang as background, but contains no occurrence of “partially bicyclic”, S_2 , or Question 3. The answer is therefore an agent-identified application rather than an explicit later resolution. It settles Question 3 in full, for every representation in the definition of (F_{*s}) ; it does not settle the other open questions in Section 6 of the source paper.

Search evidence. The four lightweight run indexes were searched for arXiv:1207.4531, the exact semigroup presentation, “partially bicyclic”, (F_{*s}) , and norm-separable weak-star fixed-point variants. Exact-pharse web/arXiv searches used the same terms. No explicit later answer to Question 3 was found. A local search of arXiv:1903.12123 located Theorem 3.1 and Corollary 3.3 and confirmed that the supporting paper cites the source but does not state this application.

Human-review recommendation. Verify the order convention $T_{st} = T_s T_t$ in the source definition and the norm-separability hypothesis in Wiśnicki's Corollary 3.3. Once those two items are checked, the displayed two-line isometry reduction is exhaustive.

References

- [1] A. T.-M. Lau and Y. Zhang, *Fixed point properties for semigroups of nonlinear mappings and amenability*, J. Funct. Anal. 263 (2012), 2949–2977; arXiv:1207.4531.
- [2] A. Wiśnicki, *Around the nonlinear Ryll–Nardzewski theorem*, Math. Ann. 377 (2020), 267–279; arXiv:1903.12123v2.